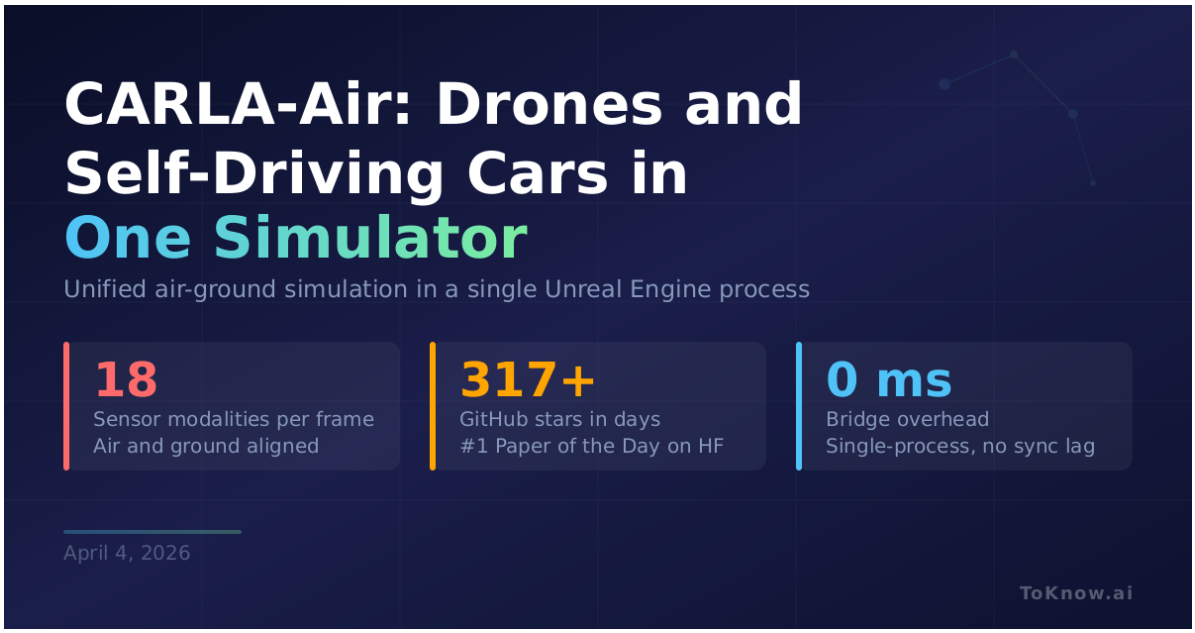
CARLA-Air: Drones and Self-Driving Cars in One Simulator, No Bridge Required

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Researchers at Peking University released CARLA-Air, an open-source simulator that puts drones and self-driving cars in the same physically consistent world. Today's tools are split: driving simulators like CARLA can't fly, and flight simulators like the now-archived AirSim can't drive through realistic cities. Connecting them via a network bridge adds 1 to 5 ms of sync overhead and breaks spatial-temporal consistency. CARLA-Air merges both into a single

Unreal Engine process: one physics tick, one rendering pipeline, zero bridge. It preserves both CARLA's and AirSim's Python APIs and ROS 2 interfaces, so existing codebases work without modification. The platform captures 18 sensor modalities per frame (RGB, depth, LiDAR, radar, IMU, and more), all aligned across air and ground agents. It sustains around 20 FPS under joint workloads and ran stable for 3+ hours with 357 spawn/destroy cycles and zero crashes.

For anyone building coordinated air-ground systems, this fills a real gap. Disaster-response teams training drones to scout while ground robots navigate, or delivery networks coordinating aerial and street-level vehicles, now have a simulator where both agents share the same rain, traffic, and physics. The project also revives AirSim's flight stack inside modern infrastructure, giving the drone research community a maintained path forward. It hit #1 Paper of the Day on Hugging Face and crossed 317 GitHub stars within days.

Simulation platforms are consolidating. Instead of domain-specific tools stitched together, the field is moving toward unified environments where heterogeneous agents coexist natively. RynnBrain showed one model handling multiple robotic capabilities; CARLA-Air extends that unification to the simulation layer itself.

Sources:

- CARLA-Air Paper (arXiv)
- CARLA-Air GitHub Repository
- Hugging Face Daily Papers
- CARLA Simulator
- Microsoft AirSim (Archived)

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